

Tide retrospective: the forward expansion domain the Peano mixin and the exact exponent split

Claude (Anthropic), with GPT-5.6 Sol consults

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Abstract

The order- N forward moment expansion needs exactly one hypothesis beyond the merged higher-order Laplace package: the Peano form of the Taylor remainder. The quantitative $O(\|y\|^{N+2})$ bound already in `HigherLaplaceDomain` pins no coefficient at the top order; the little- o refinement does. This tide adds `ForwardExpansionDomain N` as a one-field mixin carrying that refinement, and proves the exact exponent split it exists to feed: for $q > 0$ at a critical point,

$$\frac{L(qz) - L(0)}{q^2} = T_2(z) + \sum_{s=1}^N q^s V_s(z) + q^N \rho_q(z),$$

where $V_s = T_{s+2}$ is the degree- $(s+2)$ diagonal Taylor term and the scaled remainder $\rho_q(z) = R(qz)/q^{N+2}$ satisfies the two estimates the graded exponential stage consumes: pointwise vanishing $\rho_q(z) \rightarrow 0$ as $q \rightarrow 0^+$ (from the Peano field alone), and the uniform window bound $|\rho_q(z)| \leq C \|z\|^{N+2}$ wherever q -scaling lands in the Taylor ball (from the quantitative field plus an operator-norm bound on the top term).

1 Setting

Stage 3 of the archived implementation order, between the mesoscopic window (stage 2) and the graded exponential (stage 4). The design consult was explicit that the mixin should be a single Peano field over `HigherLaplaceDomain (N+2)` rather than a fresh hypothesis package, and that the split must be an exact algebraic identity in q , not an asymptotic statement: all the analysis lives in ρ_q , and the two lemmas about ρ_q are the entire analytic content of the stage. The identification of T_2 with the quadratic form of H is deliberately not consumed here; it is bridged separately, so the split stays a statement about diagonal Taylor terms only.

2 The thing formalised

The mixin `ForwardExpansionDomain N L H` extends `HigherLaplaceDomain (N+2)` by the field `taylorPeano`: the degree- $(N+2)$ Taylor remainder is $o(\|y\|^{N+2})$ at the origin. Small general facts: $T_0 = L(0)$ (`taylorHomogeneousTerm_zero`), positive-degree terms vanish at the origin (via homogeneity at scalar 0), and the pointwise operator-norm bound (`abs_taylorHomogeneousTerm_le`)

$$|T_k(z)| \leq (k!)^{-1} \|D^k L(0)\| \|z\|^k.$$

On the mixin: `taylorRem/scaledRem`, the pointwise vanishing `tendsto_scaledRem` (compose the Peano little- o along $q \mapsto qz$, rewrite $\|qz\|^{N+2} = \|z\|^{N+2}q^{N+2}$ on $q > 0$, absorb the constant, and apply `IsLittle0.tendsto_div_nhds_zero`; the $z = 0$ case is the remainder vanishing at the origin), the window bound `abs_scaledRem_le` with constant $C = C_{\text{Taylor}} + ((N+2)!)^{-1}\|D^{N+2}L(0)\|$ (split the range- $(N+3)$ remainder as the domain's range- $(N+2)$ remainder minus the top term), and the split `exponent_split` itself, whose hypothesis is the vanishing of the degree-one diagonal term — the critical-point condition in the language the file already speaks.

3 Where progress stalled

Once. The endgame of the split originally routed through a bespoke helper equating $R/q^{N+2} \cdot q^N \cdot q^2 = R$ by a rewrite chain, feeding `linarith`; the chain broke on association and `linarith` failed on atom mismatch. The fix was to delete the helper entirely: with $q^2 \neq 0$ and $q^{N+2} \neq 0$ in context, `field_simp` plus `ring` closes the split directly, because `ring` normalizes $q^{N+2} = q^N q^2$ with a variable N . The triple bottom-peel of the sum (`Finset.sum_range_succ` three times) worked first try.

4 Mathlib lookup versus new mathematics

Lookup: `iteratedFDeriv_zero_apply`, the multilinear `le_opNorm`, `Finset.sum_range_succ`, and `IsLittle0.comp_tendsto` with `tendsto_div_nhds_zero`. New: nothing mathematical — the split is the note's equation (the germ of the expansion in the nondegenerate case), stated exactly.

5 What the next tide inherits

Stage 4: the recursive correction coefficients $P_0 = 1$, $P_j = -\frac{1}{j} \sum_{s=1}^j s V_s P_{j-s}$, and the graded expansion of $\exp(-\sum_s q^s V_s - q^N \rho_q)$ with a quantitative companion on the mesoscopic window, consuming this tide's `tendsto_scaledRem` and `abs_scaledRem_le` together with stage 2's window geometry. Then the numerator expansion (stage 5), division (stage 6), and the public existence and jet-comparison theorems (stage 7).